

Assignment 5

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Electricity and Electronics EBN 122

26 September 2023

Test Information

Maximum marks:	23	Full marks:	23
Due Date:	3 October, 23:59	Open/Closed book:	Open
Total number of pages (this page included)		11	

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Assistant Lecturers: N.-J Lord, M. Malesela, G. Neeuwfan

Important

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

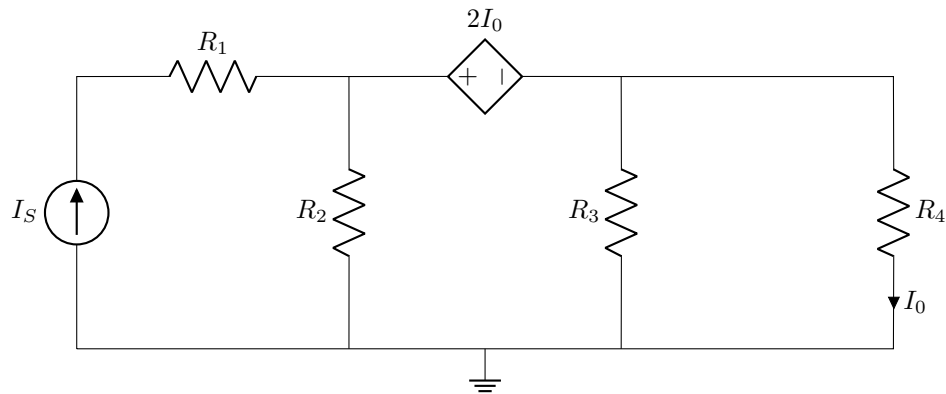
I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

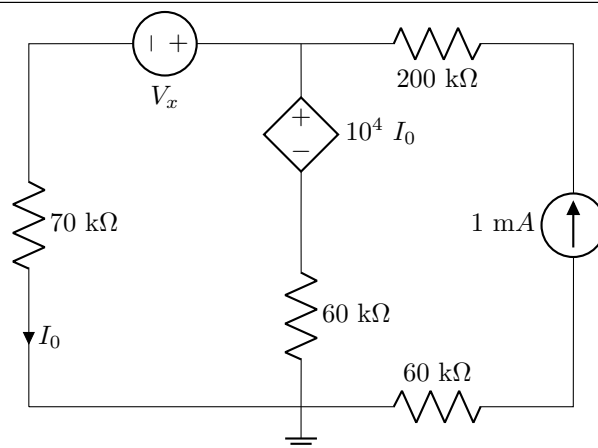
6. Contact ebn2023.consult@gmail.com **before** the deadline for assistance with technical difficulties.

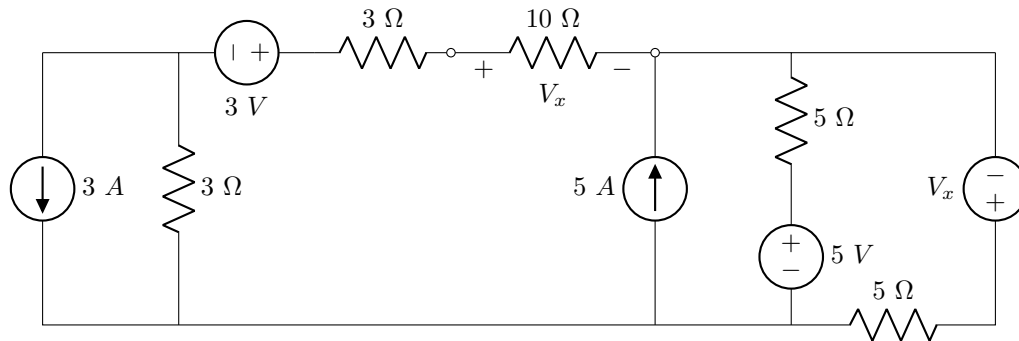
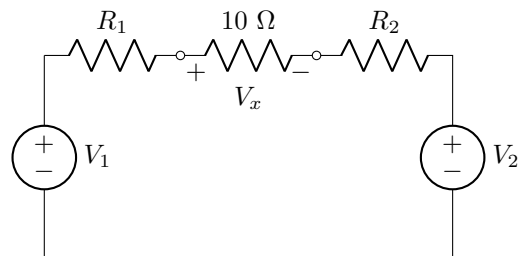
Additional guide for Complex numbers

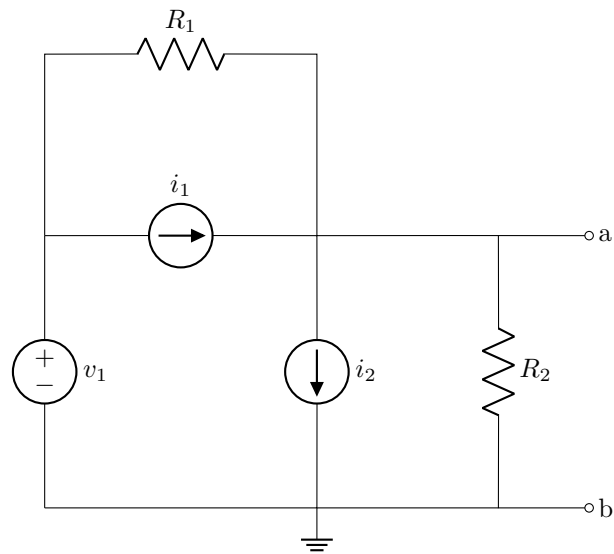
The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

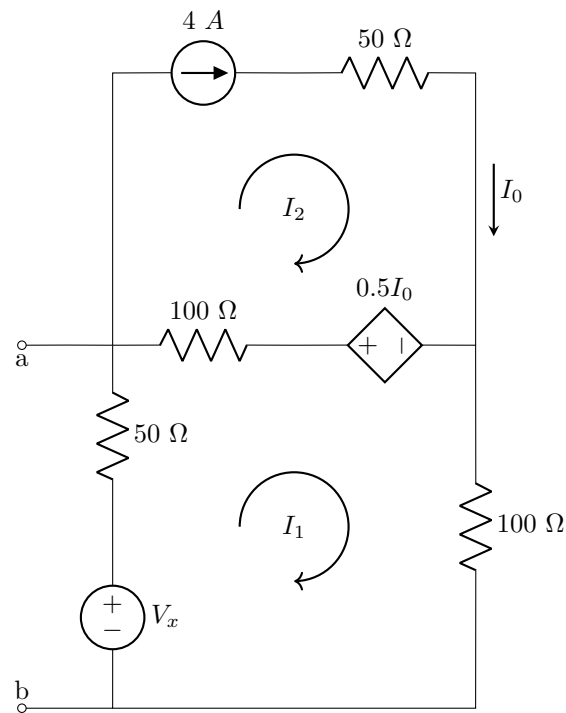
- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

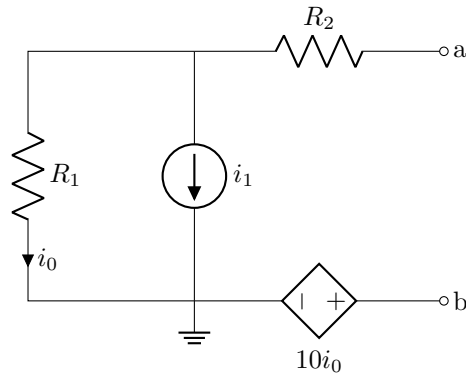
Section 1**Question 1 [2]**Given: R_1 , R_2 , R_3 , R_4 .**01** Use the principle of linearity to determine I_0/I_S . [2]

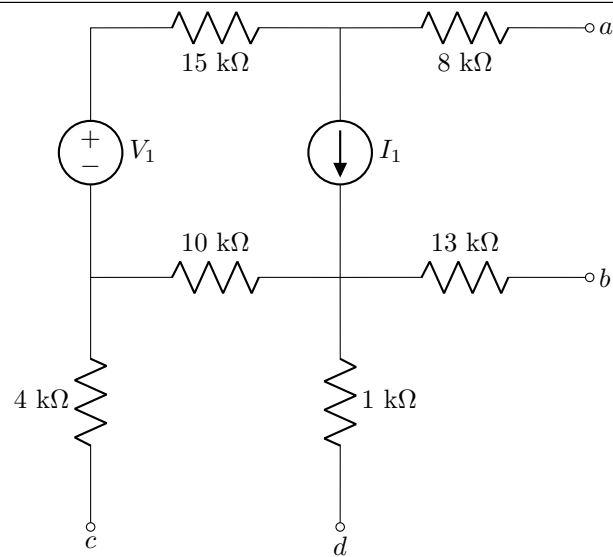
Question 2 [2]Given: V_x .**02** Find $I_o(V_x)$, the independent contribution of V_x to I_o [2]

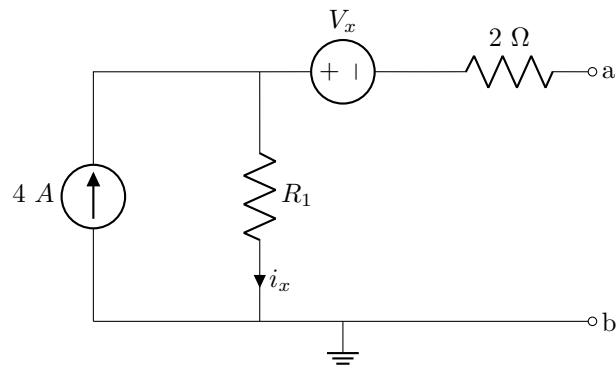
Question 3 [2]Given: V_x , V_1 , R_1 , R_2 .Use source transformation to obtain Circuit B so that it is equivalent to Circuit A. Determine V_2 in Circuit B.**Circuit A:****Circuit B:****03** | V_2 [2]

Question 4 [2]Given: R_1 , R_2 , v_1 , i_1 , i_2 .**04** Find the Thevenin equivalent voltage V_{Th} with regards to terminals a - b . [2]

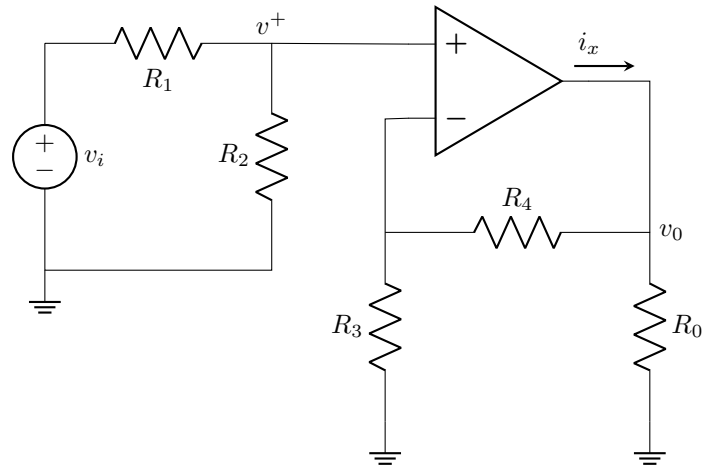
Question 5 [2]Given: V_x .**05** Find the Thevenin equivalent voltage V_{Th} between terminals a - b . [2]

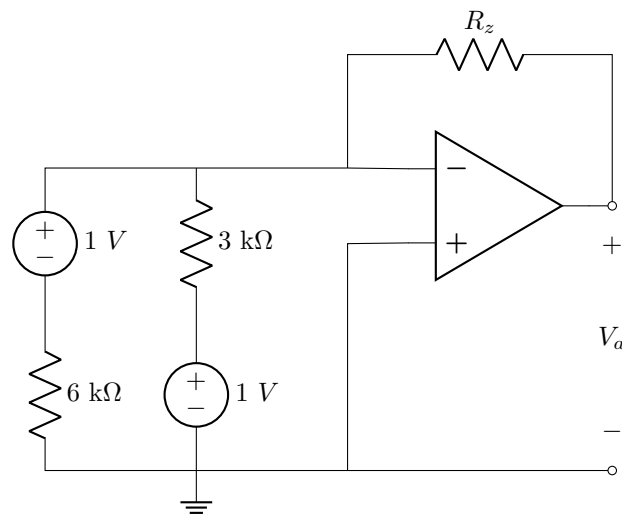
Questions 6 to 7 [4]Given: R_1 , R_2 , i_1 .**06** Determine the Norton equivalent current I_N . [2]**07** Determine the Norton equivalent resistance R_N . [2]

Question 8 [2]Given: V_1 , I_1 .**08** Determine the Norton equivalent current I_N between terminals c - d . [2]

Questions 9 to 11 [4]Given: R_1 , V_x .

09	Find the Thevenin equivalent voltage V_{Th} across terminals a - b . [2]
10	Find the Thevenin equivalent resistance R_{Th} across terminals a - b . [1]
11	Calculate the maximum power that can be transferred to a load resistor R_L that can be connected between terminals a - b . [1]

Questions 12 to 13 [2]Given: R_3, R_4 .Given: R_0, v^+ **12** Find the current i_x . [1]Given: $R_1 = R_2$ **13** Calculate the closed-loop gain (v_0/v_i) of the circuit. [1]

Question 14 [1]Given: R_z .**14** Determine V_a . [1]**Total Section 1: [23]****Grand Total : [23]**